

ECON345 Homework 2: Data Manipulation and the Tidyverse

Due: September 10th, 11:59 PM

Part I: Wrangling the Unemployment Data

In lecture we used `mutate()`, `filter()`, `arrange()`, `group_by()`, and `summarize()` on `unemp_df`. This part asks you to put all five to work end to end. This time, though, there's no CSV to import — you'll pull the unemployment data straight from FRED with `fredr`, exactly as you did in Part II of Homework 1.

Question 1 (2 pts)

Set up and pull the data from FRED. Open a new R script in Posit Cloud. Load `tidyverse` and `fredr`, set your API key from Homework 1, and pull the **monthly** `UNRATE` series from 1948 through the end of 2022 into an object called `unemp_df`.

```
library(tidyverse)
library(fredr)

# You only need to set your key once per session.
fredr_set_key("YOURKEYHERE")

unemp_df <- fredr(
  series_id      = "UNRATE",
  observation_start = as.Date("1948-01-01"),
  observation_end   = as.Date("2022-12-31")
)
```

Heads up: FRED's columns are **not** `Year` and `Unrate`. As we saw in lecture, `fredr` returns a date column and a value column, and the observations are **monthly**, not annual. Keep both facts in mind for the rest of Part I.

Question 2 (3 pts)

Bucket by decade. Use `mutate()` to create a new column called `decade`. Because FRED gives you a monthly date rather than a `Year` column, pull the year out of the date first — `(year(date) %/% 10) * 10` — following the same `%/%` trick from lecture. Assign the result back to `unemp_df`.

Question 3 (5 pts)

Summarize by decade. Using `group_by()` and `summarize()`, compute the following for each decade (remember, the unemployment rate lives in the `value` column):

1. The average unemployment rate
2. The maximum unemployment rate
3. The minimum unemployment rate

Save the result to a new object called `decade_summary`. Then use `arrange()` to sort `decade_summary` from **highest** to **lowest** average unemployment. (Because the data is monthly, each of these is computed across every month in the decade.)

Question 4 (3 pts)

Interpret. In 2–3 sentences: which decade had the worst *average* unemployment? Was it also the decade with the single worst *month*? If not, what does that tell you about the difference between a decade's average and its worst moment?

Question 5 (2 pts)

One more filter. Using `unemp_df` (not `decade_summary`), write a single piped chain that returns every month from 1970 onward where unemployment exceeded 7%, sorted from worst to best.

Part II: Reshaping the Population Data

Homework 1 introduced `decade_pop.csv` — Black and white population counts by decade, continuing the Du Bois thread from lecture. That file is **wide**: Black and White are separate columns. In lecture, we reshaped it to **long** format with `pivot_longer()` so that `group_by()` would work properly. This part asks you to redo that reshaping yourself and go a step further.

Question 6 (3 pts)

Reshape. Import `decade_pop.csv` into an object called `pop_df`. Use `pivot_longer()` to create a long-format version called `pop_long`, with a `race` column and a `population` column, exactly as shown in lecture.

Question 7 (4 pts)

Compute shares. Using `pop_df` (the **wide** version, not `pop_long`), use `mutate()` to add two new columns:

1. `total`, the sum of Black and White for that decade
2. `black_share`, the Black population divided by `total`

What happened to `black_share` between 1790 and 2020? Did it rise, fall, or stay roughly flat? (One sentence is fine — just look at the numbers.)

Question 8 (3 pts)

Growth by group. Using `pop_long`, use `group_by()` and `summarize()` to compute, for **each** race, the population in the earliest year (1790) and the population in the most recent year (2020). Which group grew by a larger raw number? Which group grew by a larger multiple (i.e., which population is a bigger number of *times* larger in 2020 than in 1790)?

Submission

Upload your `.R` script to Canvas before the deadline. As always: comment your code, and if you get stuck, show your work — partial credit exists for a reasonable attempt, not just a correct final answer.